Instructor: Dr. Jim D. Pham and Paul Nguyen
Email: jpham@ohlone.edu
Office Hrs: Thursday, 7 PM – 8PM (Canvas)
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Section: CNET-172-1 (100% online)

COURSE DESCRIPTION:

CCNA Security equips students with the knowledge and skills needed to prepare for entry-level security specialist careers. CCNA Security aims to develop an in-depth understanding of network security principles as well as the tools and configurations required to secure a network. Prepares students for the Cisco CCNA-Security certification exam.

Textbook (Optional):

CCNA Security Course Booklet Version 2
By: ciscopress.com
ISBN: 978-1-58713-351-0

CCNA Security Lab Manual Version 2
By: ciscopress.com
ISBN: 978-1-58713-350-3

STUDENT LEARNING OUTCOMES:

After completing the course, students will be able to:

1. Describe the security threats facing modern network infrastructures.
2. Secure network device access by implementing AAA, using ACLs, and secure network management and reporting.
3. Mitigate common Layer 2 attacks.
4. Implement the Cisco IOS firewall feature set, an ASA, the Cisco IOS IPS feature set,
5. and site-to-site IPSec VPNs.
6. Administer effective security policies.

METHODS OF EVALUATION:

1. Objective quizzes on security threats facing modern network infrastructures.
2. Lab Projects to practice skills in securing network device access by implementing AAA,
3. using ACLs, and secure network management and reporting; mitigating common Layer 2 attacks; and implementing the Cisco IOS firewall feature set, an ASA, the Cisco IOS IPS feature set, and site-to-site IPSec VPNs.
4. Comprehensive Final Exam on network security in a Cisco network environment, and preparation for the Cisco Certified Network Associate with Security (CCNA-Security) certification examination.
5. Skills-based assessment (hands-on final exam) on CCNA-Security steps, practices, and techniques.

TEACHING METHODS:

1. This course is a hands-on, career-oriented e-learning solution that emphasizes practical experience.
2. Lectures: The presentations slides and assignments will be accessed through Ohlone's Canvas Course Management System. You access the online resource at this URL: https://ohlone.instructure.com/
3. Discussion boards for students to participate in discussion threads and collaborate with other students in the class.

DISCUSSION BOARD:

Your participation on the Discussion Board (on Canvas) will be an important aspect of the class. You will be required to post, and respond to at least two posts from a classmate, on a weekly basis.

INSTRUCTIONAL RESOURCES:

1. **NetSpace –** Cisco Network Academy Site online learning environment (the textbook) - provides access to the CCNA Security Course Booklet/Lab Version 2 course content. https://www.netacad.com You will need a separate account for this site. If have completed any of the Cisco Networking courses you may use that same account.
2. **Canvas:** Class documents, syllabus, discussion, and the lecture slides will be posted on Ohlone's Canvas. Canvas is an online Course Management System. You access Canvas at this URL: https://ohlone.instructure.com.

COURSEWARE:

CCNA Security Course Booklet/Lab, web-based course provides CCNA Security. The course includes curriculum content, lab exercises, and assessments within an engaging, hands-on learning environment. For an overview of the course and course content use the following link: https://www.netacad.com/group/resources/ccna-security/2.0. The CCNA Security is accessible through the Cisco Network Academy NetSpace.

CANVAS CONFERENCES:

I will be conducting weekly live lectures and labs using Canvas Conferences

EMAIL:

If you send me email, please put **course number – name – topic** into the email **Subject** field.
For example: CNET 172 – Name – Issue

LAB ASSIGNMENTS:

There will be Packet Tracer activities and lab activities each session.

POLICY ON STUDENT WORK:

The assignments are to be done individually, unless a group project is specified. You are encouraged to talk to fellow students about concepts and methods, but you may not copy someone else's work. I adhere to the **Ohlone College Policy on Academic Integrity**.

QUIZZES:

There will be chapter exams and quizzes on the reading assignments and on the prior week's material. The quiz covers the content in the reading assignments and on the prior week's material. It is designed to provide an additional opportunity to practice the skills and knowledge presented in the chapter and to prepare for the Chapter Exam. You will be allowed multiple attempts and the grade does not appear in the gradebook.

GRADING:

Labs:	50%
Participation:	10%
Chapter Exams:	10%
Final Exam:	30%

GRADING SCALE:

A = 90-100
B = 80-89
C = 65-79
D = 55-64
F = <55

COURSE SCHUDULE:

The following is a recommended schedule. The instructor reserves the right to make any schedule changes deemed necessary.

Session	Date	Activity
1	**8/28/17**	**Download instructional materials** - Course Introduction - First Time in This Course - Student Support and Resources - Download software for the class - Pretest Exam **Due date:** 09/04/2017
2	**8/31/17**	**Chapter 1: Modern Network Security Threats** - Launch Chapter 1 - Chapter 1 Exam - Chapter 1 Lab Activities - Lab Activities for this Chapter **Due date:** 09/11/2017
3	**9/4/17**	**Holiday: Labor Day** (Ohlone College closed)
4	**9/05/17**	**Chapter 2: Securing Network Devices** - Launch Chapter 2 - Chapter 2 Exam - Chapter 2 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 09/14/2017
5	**9/11/17**	**Chapter 3: Authentication, Authorization, and Accounting** - Launch Chapter 3 - Chapter 3 Exam - Chapter 3 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 09/18/2017
6	**9/14/17**	**Chapter 4: Implementing Firewall Technologies** - Launch Chapter 4 - Chapter 4 Exam - Chapter 4 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 09/21/2017
	9/18/17	**Chapter 5: Implementing Intrusion Prevention** - Chapter 5 Exam

		- Chapter 5 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 7/18/2017
7	**9/21/17**	**Chapter 6: Securing the Local Area Network** - Chapter 6 Exam - Chapter 6 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 09/25/2017
8	**9/18/17**	**Chapter 7: Cryptographic Systems** - Chapter 7 Exam - Chapter 7 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 09/28/2017
9	**9/25/17**	**Chapter 8: Implementing Virtual Private Networks** - Launch Chapter 8 - Chapter 8 Exam - Chapter 8 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 09/02/2017
10	**9/28/17**	**Chapter 9: Implementing the Cisc- Adaptive Security** - Launch Chapter 9 - Chapter 9 Exam - Chapter 9 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 10/05/2017
11	**9/02/17**	**Chapter 10: Advanced Cisco- Adaptive Security Appliance** - Launch Chapter 10 - Chapter 10 Exam - Chapter 10 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 10/09/2017
12	**10/05/17**	**Chapter 11: Managing a Secure Network** - Chapter 11 Exam - Chapter 11 Lab Activities - Complete Packet Tracer Activities and Lab Activities for this Chapter **Due date:** 10/12/2017
13	**10/09/17**	**Practice Exams** - CCNA Security 2.0 PT Practice SA Part 1 - Practice Final Exam **Due date:** 10/16/2017
14	**10/12/17**	**Practice Exams cont.** - CCNA Security 2.0 PT Practice SA Part 2 - Practice Final Exam **Due date:** 10/20/2017

15	**10/16/17**	**Hands On Skills Exam** - Hands On Skills Exam - Submit configuration of devices via Canvas - Course Feedback **Due date:** 10/20/2017
	10/19/17	**Hands On Skills Exam (Cont.)**
	10/20/17	**Final Exam**